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Accelerating Sustainability and Decarbonisation Within the Offshore Energy Supply Chain

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Logistics emissions from freight and warehousing account for at least seven percent of global greenhouse gas emissions, according to [McKinsey](#), and within marine operations and logistics, there is a plethora of challenges and considerations to ensure businesses are operating in a way which minimises environmental impact through solutions which are not only commercially viable but commercially attractive.

Peterson Energy Logistics is committed to the decarbonisation of the offshore energy supply chain in marine logistics, through collaboration with our customers via a variety of strategic initiatives and innovations such as pooling, digitalisation, decarbonising assets, and minimising our footprint through carbon neutral practices.

In this paper, we will share our experiences decarbonising our own operations, as well as how we both deliver and empower practices which directly impact our customers' ability to drive their own decarbonisation agendas.

Introduction

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Peterson offers fully carbon neutral services and invests in innovations which not only cut our carbon footprint but drive new and more efficient ways of working that are futureproofing our business, as well as that of our customers.

Energy logistics can be a carbon intensive practice – ensuring the safe, timely, cost-effective, customer-focussed, responsive, flexible, and strategic movement of goods critical to the success of the offshore energy sectors. However, we have shown that sustainability, decarbonisation, and future solutions practices can add to, rather than come at the cost of, providing world-class, cost-competitive services, and that there is far more that can be achieved through enhanced collaboration across the Middle East.

Peterson Energy Logistics is a global lead logistics partner, with established bases in the UK and Netherlands, as well as providing expert services in the Middle East and Asia Pacific regions. Alongside our physical logistics offering which covers an array of services from the cranes on quaysides to materials management, warehousing, vessel management, road transport, and aviation, we also have strong technology and consultancy products and expertise which are furthering the impact of our role in the decarbonisation agenda on a global scale.

We are both a consultancy partner and an implementation partner through deploying technology, redesigning logistics networks, and supporting operators to embed sustainable practices across marine operations. As well as offering strategic guidance, we are capable of delivering real-world change.

Peterson Energy Logistics is part of Royal Peterson and Control Union which employs 7000 people across 80 countries offering services including control, testing, and inspections, as well as logistics, across diverse sectors.

Collaboration in the Middle East

The Gulf region has invested heavily in net zero efforts and remains ambitious and committed to meaningful progress. [Seatrade Marine](#) reported that Abdulla Bin Damithan, CEO & MD, DP World GCC said: "We can't stand still – even if the odds seem against us. Regulatory frameworks are already shaping how vessels operate and where they berth. We have no choice but to decarbonise – and fast." His comments came at this year's [Seatrade Maritime Logistics Middle East](#) conference, where H.E Suhail Mohamed Al Mazroui, Minister of Energy & Infrastructure, UAE MOEI, further outlined the commitment to decarbonisation and the net zero 2050 target.

In its recent white paper, Achilles cited that the region has some of 'the world's most ambitious infrastructure and industrial development underway' and must balance that growth alongside managing the carbon footprint of the supply chain. It reports that Scope 3 emissions in construction, energy, and manufacturing often exceed 70% of all emissions.

Through our experiences, we have seen a desire to harness learning from other parts of the world but to tailor it and deliver in a smarter way. There are fantastic opportunities for collaboration – exporting the lessons, solutions, and best practices which we have learned through decades in the North Sea and across the world, and apply them, along with a culture of innovation, to projects and customers.

Due to the speed of growth in the region, companies are now trying to streamline processes, wrangle data, and lay the foundations of operations to be ready for the next wave of opportunities. The Middle East feels incredibly progressive with a huge appetite and curiosity for digitalisation and future solutions at pace and scale which support business growth. This future includes a commitment to decarbonisation which is a growing priority in the Middle East, underpinned by regional commitments including Saudi Vision 2030 and UAE Net Zero 2050 placing pressure on supply chains to explore their role in achieving these goals.

Our energy management journey

We have taken significant steps to lead the way in decarbonisation through a range of initiatives, offering our customers fully carbon neutral operations. This not only cuts our impact as a business but supports our customers' Scope 3 reporting. Undertaking this journey within our own business also bolsters our insight, expertise, and knowledge when providing strategic consultancy and implementation procedures to our customers who work with us to shape their decarbonisation roadmaps.

We have established and verified systems to measure, manage, and offset emissions effectively ensuring our carbon neutrality claims are credible, and that sustainability is embedded across operations. We actively collaborate with clients to measure and report their share of Scope 3 emissions within our operations, ensuring transparency and driving mutual progress toward net zero. Sustainability considerations are woven throughout our operational and strategic decision making.

This is not a given in the logistics industry, [McKinsey](#) reports that half of companies have no decarbonisation goals in place and only a quarter have both set goals and believe they have the means to achieve them, with a similar number believing they will be unable to reach their carbon reduction targets.

Our energy management journey began in 2017 with a goal to reduce Scope 1 and 2 emissions by 6% by 2022 across all ISO 50001-certified sites. We exceeded this, achieving a 39% reduction in energy consumption and 47% decrease in carbon emissions. In 2022, we became one of the first energy logistics company to achieve PAS2060 carbon neutrality status and in April 2025 became the first company to achieve ISO14068 accreditation through NQA. The standard provides a rigorous, transparent framework for measuring, reducing, and offsetting carbon emissions.

Stephen Burt, Carbon & Sustainability Services Director at NQA, said: "As their Verifier, I was very pleased to complete NQA's first UKAS Accredited Verification for ISO 14068-1 with Peterson Energy Logistics. ISO 14068-1 is a powerful tool in making the transition to net zero, and Peterson Energy Logistics are leading the way. I was very impressed with the company's commitment to and performance in decarbonisation using industry leading techniques to progress them towards their net zero target."

Within our own operations we have implemented a range of measures including powering our buildings through renewables electricity contracts, electrifying our operational fleet of vehicles, using HVO100 fuel where electrification has not been possible, local procurement prioritisation, waste reduction, and employee engagement.

As an industry leader in this space, we are committed to sharing our learning and experiences with customers and wider stakeholders to empower the wider decarbonisation of our industry.

Introduction to Lighthouse

In the academic paper *Can supply chain digitalization open the way to sustainable development?* Evidence from corporate ESG performance, researchers stated that "supply chain digitalization significantly enhances corporate ESG performance" by "enhancing internal operational efficiency, increasing interfirm trade credit, and enhancing external supervision." It also states that "the promotion effect of supply chain digitalization on ESG performance is mainly present in corporates with stronger cooperative culture, stronger innovation capabilities, party organization, and those operating in heavily polluting industries." This point about heavily polluting industry is of interest when considering the energy industry's role in decarbonisation of oil and gas as a vital lever in achieving a just energy transition.

Peterson has been leading the way in the innovation of the energy logistics industry for more than a decade though its proprietary technology suite, Lighthouse, which illuminates the end-to-end logistics supply chain through apps which plan, track, and analyse areas including quayside management, transport, port activity planning, CCU fleet management, vessel activity, and cost allocation. Decarbonisation is achieved through outcomes including the consolidation of cargo, adherence to planned activity schedules, reduction in non-productive time, effective route planning, and effective cargo management.

Lighthouse is an ecosystem of applications which integrate together, as well as with third party systems, bringing together data for detailed insight.

We first invested in Lighthouse around 15 years ago and our operational teams have worked hand-in-hand with our technology team to deliver and continuously refine a system which delivers results that end users need from their data systems.

Benefits of Lighthouse

Within its Transport app, an optimiser tool automates load planning, maximising cargo per vehicle, and ensuring timely delivery. This reduces unnecessary trips, cutting emissions, and improving logistical performance while removing the burden of a time-intensive process. This has clear benefits in terms of enhancing sustainability in the carbon-intensive practice of road freight forwarding.

In terms of the role it plays in identifying and delivering efficiencies, for a client in the Middle East, Lighthouse has streamlined operations, reduced asset down time, and improved vessel deck utilisation. This has led to a 20% reduction in the vessel fleet and a 22% reduction on CO2 emissions - the equivalent of 8,915 tonnes or approximately 250,000 trees every year.

For Peterson, the investment in Lighthouse represents an important step in modernising traditional practices and holds further potential for efficient, informed, strategic value to our customers.

As well as being an operational tool which benefits Peterson's logistics customers, Lighthouse is also available as a Software as a Service model whereby businesses work with our expert team to select the apps which best meet their requirements in terms of operational, strategic, and decarbonisation objectives.

Lighthouse and AI

Peterson has recently invested in deploying artificial intelligence and large language models into Lighthouse to enhance its ability to be a strategic value add to customers. Not only is AI streamlining many day-to-day tasks and analysing, at scale, opportunities, challenges, operations, and potential solutions at the click of a button, it is taking on a lead role in logistics modelling drawing upon the system's 10m tonnes of cargo process which provides as a significant data set for analysis and benchmarking. Through technology, customers are making informed decisions about decarbonisation, efficiencies, operational priorities, and strategic developments.

Pooling and sharing

In a resource heavy environment, Peterson strongly believes and leads the way in collaboration as a means of achieving efficiencies. Through pooling and sharing, customers achieve not only carbon efficiencies, but a reduction in costs, enhanced processes, and better operational outcomes. This is not a theoretical opportunity but something we've delivered successfully across the globe and see significant potential for other operators in the Middle East.

Peterson is the manager and facilitator of the Southern North Sea (SNS) Pool from the Netherlands. This fully shared marine logistics supply chain has increased utilisation and efficiency of assets leading to a 36% reduction in emissions. Servicing 150 offshore locations for nine operators, it consolidates platform supply vessel (PSV) activities across operators into a single, optimised schedule, reducing duplication, and maximising vessel utilisation. It currently operates a fleet of 11 PSVs, flexibly deployed based on shared demand.

The Pool's operations are underpinned by sustainability through a range of initiatives including cleaner fuel use for vessels, electric onshore vehicles, and energy efficient logistics hubs. Peterson offers a comprehensive suite of shared onshore services, including warehousing, cargo handling, helicopter operations, and equipment maintenance. This integrated approach creates additional efficiencies and cost benefits for customers.

Similar shared logistics models operate with six to 10 vessels in Aberdeen and up to 18 vessels in the Ymond Pool. Together, these examples prove the scalability and flexibility of this collaborative approach.

In the Middle East, we established a joint marine and port logistics operations centre for a customer. We assessed our client's operating model, leading to the development of a strategic sharing framework. A comprehensive roadmap with risk strategies and contingency plans were delivered and effectively executed

for a vessel sharing model which has created the conditions for \$18m savings and 28% OPEX savings per annum.

These are practical, proven approaches that lead to measurable emissions reductions which have been proven to be scalable and replicable globally. Pooling strategies and operational optimisation are critical enablers of decarbonisation and are a prime method for Middle East operators to achieve their carbon reduction goals.

Decarbonisation of fleet

As part of a global shift towards more sustainable logistics, especially in fuel heavy sectors, Peterson has actively sought and embedded solutions to cut emissions without compromising operations. Peterson has made significant progress in decarbonising our vehicle fleet. While our focus remains on electrifying as much of our fleet as possible, we recognise the energy transition is a journey and have adopted fuel additives and alternatives to better our environmental performance.

Half of our transport fleet has been converted to using Hydrogenated Vegetable Oil (HVO) 100. HVO100, known as renewable diesel, is made from renewable resources like used cooking oils, animal fats, and non-food grade crops and leads to a significant reduction in greenhouse gas emissions compared to fossil diesel. The use of HVO can reduce greenhouse gas emissions by up to 90% over its lifecycle when compared to using conventional diesel which many of our trucks previously used.

Seagoing vessels were identified as our biggest direct source of CO₂ emissions. With no low emission fuel alternatives widely available, we decided to test fuel additives as a practical, cost-effective step. Marine Gas Oil (MGO) is currently the primary fuel for powering Platform Supply Vessel (PSV) fleets and combining a fuel enzyme with MGO is proving to reduce fuel consumption and carbon emissions by 10-15%.

Enzyme Fuels

Enzyme fuel additives such as XBEE have led to impressive environmental and financial efficiencies – first through the SNS Pool managed from Den Helder in the Netherlands, before being trialed with a client in the Middle East in offshore support vessels. The enzymes - natural biocatalysts derived from plant leaves - enhance fuel combustion by stabilising hydrocarbons, reducing contaminants, and gradually breaking down carbon build-up. This boosts energy conversion, extends engine life, and reduces maintenance. In addition, the enzyme cleans the entire fuel system, from storage tank to engine, preventing problems with microorganisms in fuels.

To facilitate the trial, Peterson Energy Logistics introduced the enzyme provider to the vessel owners and answered all technical questions. The Peterson fuel team led the project, providing the enzyme and working closely with the operator and vessel owners to agree a method of dosing and delivery that met the technical requirements of the vessels and safety handling needs. Procurement, customs, delivery, and safety controls were all managed in-house by the Peterson team, and the vessels used independent management systems to gather and analyse the data, ensuring all results recorded were accurate and robust.

Netherlands savings

In the Netherlands, use of this enzyme has led to a 11% reduction in fuel use, 3000 tonnes reduction in CO₂, and 16% reduction in noxious gases from our SNS fleet.

A six-month trial, from Aberdeen in the UK as a base, saw fuel savings of up to 15%, equating to savings of over 290m³ of MGO per vessel per annum, which has a value of over £200,000, resulting in a reduction of over 800 tonnes in CO₂ emissions. It was estimated that should all operators in the Northern North Sea opt to use this enzyme, 292,000 tonnes of carbon could be removed from the marine supply chain.

Middle East savings

In the Middle East, working with a customer with a hot, high-load, and consistent offshore environment, fuel consumption was reduced by almost 10% and NOx emissions reduced by more than 18%.

The success of this trial in the Middle East has sparked interest from other offshore operators, as well as onshore industrial and agricultural companies across the region. Peterson is now working with these organisations to explore how fuel enzymes can reduce emissions and operating costs in their respective sectors.

Consultancy

McKinsey reports that 40 to 50% reduction in logistics emissions could be achieved through solutions available today through initiatives such as supply chain efficiency, technology levers, transport, and warehousing with measures such as optimising routes and loads, electrification of fleet, and building efficiencies. These are all measures we are embracing at Peterson to decarbonise our operations and reduce our customer's Scope 3 emissions. And we bring this mindset and experience to our consultancy services to support others on their journey and apply learning and innovation to their unique contexts.

At Peterson, we bring deep industry expertise in energy, resources, and heavy industries, coupled with a practical, operations-focused approach to consulting. We provide emission analysis and reduction strategy; supply chain and procurement design through a carbon lens; and stakeholder engagement, technology and reporting.

A dedicated consulting team leads the entire discovery, delivery, and analysis process, working with a global network of subject matter experts in areas including value chain diagnostics, technical solutions, greenfield supply chain solutions, ESG strategy, and decarbonisation. Our role is to help clients reduce emissions at source by designing and operating more efficient supply chains. That is where the biggest gains can be made, both now and in the future. Whether working with existing operations or supporting new developments, our emphasis is on eliminating inefficiency and waste through better design and smarter execution.

We empower transformation through cutting-edge innovation, driving unparalleled value through optimised, safe, sustainable, and efficient supply chain solutions which achieve tangible business outcomes.

Our team has been engaged in the development of green ports including carbon footprint calculation and reporting which is enabling sustainable port operations and environmental compliance.

Middle East consultancy

Within oil and gas, supply chain optimisation has a huge role to play in decarbonisation. Peterson analysed and optimised the supply chain of a major operator in the Middle East region. The company was operating on an outdated logistics model resulting in inefficient processes.

Working with our consultants on a supply chain logistics optimisation study and use of Lighthouse to implement and embed more efficient processes and digitise their supply chain, they have benefited from a significant 32% reduction in logistics costs and 22% reduction in carbon emissions. There have also been improvements in vessel departure compliance, cargo inspection compliance, deck utilisation, CCU utilisation, reduction in incidents and non-compliances, and improvements in fuel consumption which all further their decarbonisation journey, alongside their commercial growth and innovation.

Conclusion

Peterson's experience globally, and practical implementation in the Middle East, is something we believe can and should be used to significantly enhance efforts of the decarbonisation, sustainability, and future solutions which will underpin the future of the marine sector, and the role logistics plays in the wider success of the region's energy industry.

This paper has detailed how data, collaboration and logistics coordination can drive meaningful emission reductions. Digital platforms provide a way for clients to report, track, and improve their emissions performance year-on-year, as well as achieve optimal planning into the supply chain to minimise the movement of goods across locations. Large scale initiatives such as pooling provide tangible results beyond decarbonisation. Smaller scale investments in alternative fuel use and more efficient energy management are also a critical part of the puzzle.

The focus on decarbonisation is vital as an environmental and social governance focus but, as a true partner of our customers, we also ensure this is aligning with wider business priorities, objectives, ambitions, and growth. As businesses in the Middle East consider environmental sustainability, decarbonisation, and future solutions at this important juncture, it is vital that they do so with a holistic approach rather than it being viewed as an add-on. The investment in sustainability should be seen as a value add now, rather than just an investment in the future.

At Peterson, we are proud of our track record of achievement in the decarbonisation of logistics for us and our customers. What excites us is how that track record drives not only our future, but how it can shape and inspire this vital facet of the energy industry in the decades to come.

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